

Reviewer Pack — Non-Abelian Harmonic Analysis and SOS Refutation Degree for ...

Entry 93e2fe31dd48 · INCONCLUSIVE

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Non-Abelian Harmonic Analysis and SOS Refutation Degree for CSPs

Entry ID: 93e2fe31dd48 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-21 01:05:36 UTC

1. Conjecture

Field A (mathematical branch): Non-Abelian Harmonic Analysis **Field B** (complexity object): SOS Refutation Degree for CSPs

Statement:

For a CSP instance over a non-abelian group G , the SOS refutation degree is bounded by the dual norm of the Fourier coefficients of the CSP's constraint function. Specifically, for every CSP instance φ over G , $\text{refutation_degree}(\varphi) \leq \|F\|_\infty$, where F is the Fourier transform of φ 's constraint function.

Rationale (proposer's reasoning):

Non-abelian harmonic analysis decomposes CSP constraints into group characters, revealing structural symmetries that may influence the SOS hierarchy's ability to certify unsatisfiability. The dual norm of Fourier coefficients quantifies the 'spread' of constraints across irreducible representations, which could correlate with the minimal SOS degree required for refutation.

Taxonomy category: SOS_HIERARCHY (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 0e7243ef883f0cff

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.95	UNCERTAIN	SAFE
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction
import sys

def generate_3sat_instance(n):
    clauses = []
    for _ in range(2 * n):
        clause = [random.choice([-1, 1]) * (i + 1) for i in range(n)]
        if not any(clause[i] == -clause[j] for i in range(n) for j in range(i + 1, n)):
            clauses.append(clause)
    return clauses

def fourier_transform(F):
    n = len(F)
    F_hat = [0] * n
    for k in range(n):
        for i in range(n):
            F_hat[k] += F[i] * (Fraction(math.cos(2 * math.pi * i * k / n), 1) + Fraction(math.sin(2 * math.pi * i * k / n), 1))
        F_hat[k] /= n
    return F_hat

def sos_refutation_degree(clauses):
    # Simplified SDP relaxation for refutation degree (basic DPLL)
    # This is a placeholder and should be replaced with actual SOS computation
    return len(clauses)

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n = 40
    instances_tested = 30
    metric_value = 0.0
    conjecture_holds = True
    counterexample = ""

    for _ in range(instances_tested):
        clauses = generate_3sat_instance(n)
```

```

F = [1] * (2 ** n) # Placeholder Fourier coefficients
F_hat = fourier_transform(F)
refutation_degree = sos_refutation_degree(clauses)

if max(abs(coeff) for coeff in F_hat) < refutation_degree:
    conjecture_holds = False
    counterexample = "SOS refutation degree exceeds max Fourier coefficient"
    break

metric_value = max(abs(coeff) for coeff in F_hat)

return {
    "metric_name": "max_fourier_coefficient",
    "metric_value": metric_value,
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    seeds = [int(s) for s in sys.argv[1:]] if sys.argv[1:] else [2, 3, 5, 7, 11, 13, 17, 19, 23, 29]
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = (sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results)) ** 0.5
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6c3843cd.py", line 74, in <module>

```

    result = run_trial(seed)
    ~~~~~^~~~~~

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6c3843cd.py", line 50, in run_trial

```

    F = [1] * (2 ** n) # Placeholder Fourier coefficients
    ~~~~~^~~~~~

```

MemoryError

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Test crashed with MemoryError before producing results, preventing evaluation of support fraction or counterexamples. | next: Optimize memory usage in test code for large n, then re-run experiments

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	86497
2	preregistration	ollama_remote	qwen3:8b	0	0	32198
3	novelty	ollama_remote	qwen3:8b	0	0	27892
4	novelty	ollama_remote	qwen3:8b	0	0	21198
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14891
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15436
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11323
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9404
9	judge	ollama_remote	qwen3:8b	0	0	21162

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 240000 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/93e2fe31dd48.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/93e2fe31dd48.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/93e2fe31dd48.tar.gz` (if generated)